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Should Gulf desalination follow the sun?

Renewable-Powered Desalination: A Sustainable Alternative to Conventional Water Treatment Systems

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UAE water security runs through desalination

Three facts from a 2025 UAE study that put this question at the centre of Theme 1.

>90%

of the UAE's drinking water comes from desalination

≈15%

of UAE electricity is used to desalinate seawater

>22%

of UAE energy-related CO_2 emissions come from desalination

Source: Nwafor et al. 2025

The obvious next step is more solar

Al Taweelah in Abu Dhabi, among the world's largest reverse-osmosis plants, already runs partly on solar (ACWA Power and EWEC 2022; Bellini 2022). The open question is how the plant should operate around it.

The standard answer: let the plant follow the sun

Dispatch studies treat the plant as a converter of power into water. The membrane is not in the model.

What dispatch studies report

>20%

cost reduction reported for scheduling an RO plant around demand response (Wu et al. 2024)

- RO load matched to variable solar supply (Zein et al. 2023; Mito et al. 2022)
- Plants bid flexibility into demand-response markets (Oikonomou & Parvania 2020)
- The most recent dispatch model leaves out membrane fouling and Gulf conditions (Alahmed et al. 2026)

What membrane experiments show

37%

of permeability lost in three days by a seawater membrane under solar-like variable operation, without rinsing (Dimitriou et al. 2025)

- A brackish plant run intermittently for 14 years lost about half its permeability (Ruiz-García & Nuez 2020)
- Gulf feedwater is harsher still: average salinity 39 psu (Omerspahic et al. 2022)

An hourly model with the membrane inside

Solar dispatch, plant energy use and membrane fouling are solved together, hour by hour, on measured Abu Dhabi weather.

1

Sun and sea

Hourly irradiance, air temperature and aerosols from NASA POWER, Abu Dhabi coast, 2023.

2,009

kWh/m^2 of sunlight over the year

2

Plant physics

Energy per cubic metre from feed salinity and temperature, with energy recovery.

3.17

kWh/m^3 in summer; a full-scale Gulf plant is projected at 3.11–3.13 (Kim & Hong 2023)

3

Membrane wear

Fouling that cleaning removes, plus permanent loss from heat, idling and every restart.

250–8,000

start/stop cycles a membrane survives, the range we test

4

Controller and cost

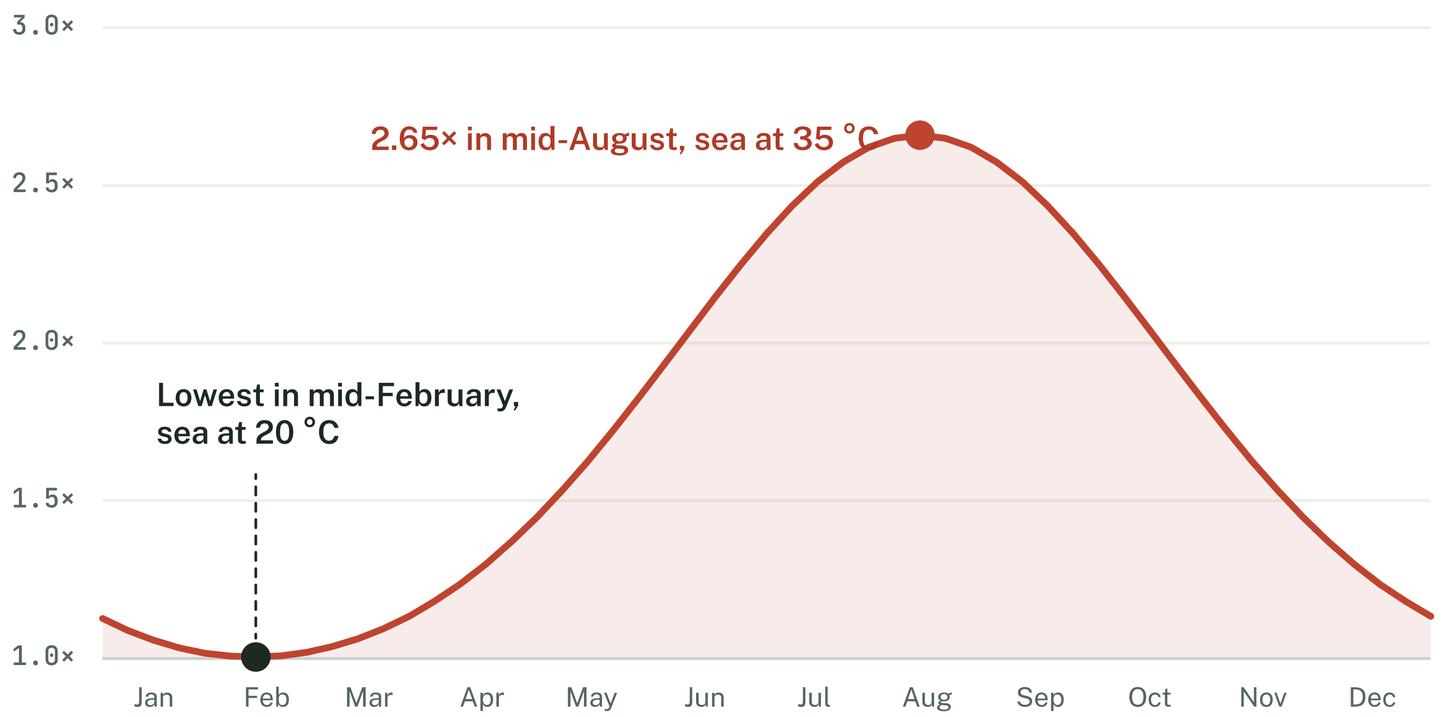
Dispatch and storage optimised for the cheapest water over the plant's life.

2

controllers on the same plant: one ignores wear, one prices it

Summer brings cheap solar and the fastest fouling

Modelled membrane fouling rate through the year on Abu Dhabi feedwater, relative to its February low.



Fouling rate, August vs February

2.65×

faster, because fouling speeds up in warm water

Plant energy, winter vs summer

3.78 / 3.67

kWh/m³: cold water passes the membrane more slowly. A model prediction; plant data on this effect are mixed (Kim et al. 2019).

Follow the sun blindly, and water costs 45% more

Same plant, same solar array. The only change is whether the controller counts membrane wear.

Renewable share	Restarts per year	Membrane life	Cost of water
52.3%	361	0.53 yr	+45%
up from 39.5%	from 0 at steady output	down from 2.06 years	0.992 to 1.434 USD/m³

Wear ignored

Follows the sun

1.434 USD/m³

361 restarts a year · membranes last 0.53 years

Wear counted

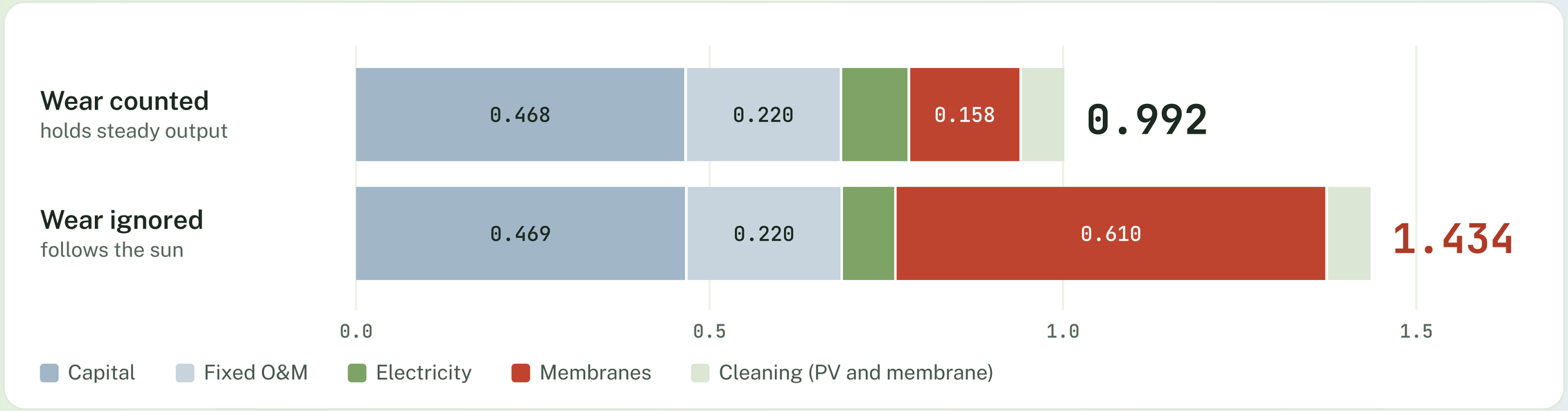
Holds steady output

0.992 USD/m³

No restarts · membranes last 2.06 years

The membrane bill is 22 times the energy saving

Cost of water by component, USD/m^3, for the two controllers on the same plant and solar array.



Totals are net of a small export revenue from surplus solar (0.014 USD/m^3 or less), so the bars run slightly past them.

Electricity

-0.021

USD/m^3 saved by running on more sunlight

Membranes

+0.452

USD/m^3 added by faster wear

Membrane life

2.06 → 0.53 yr

replaced about four times as often

The number that decides it

4,200

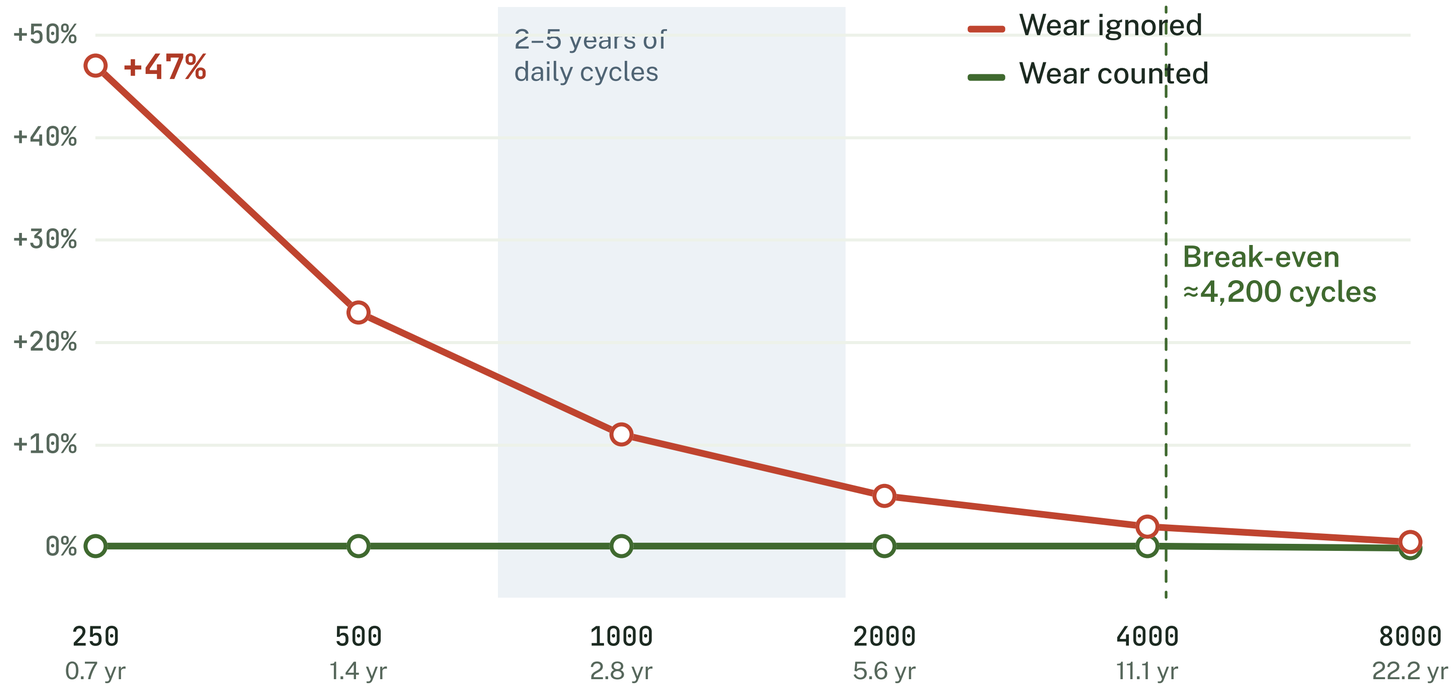
start/stop cycles a membrane must survive before following the sun pays



Central case, biofouling index 2. Replacement range: Avlonitis et al. 2003.

Flexibility can gain 0.2% or lose 47%

Change in the cost of water versus steady output, across membrane cycle tolerance. Central case, biofouling index 2.



Worst case, wear ignored

+47%

cost of water at 250 cycles

Best case, wear counted

-0.2%

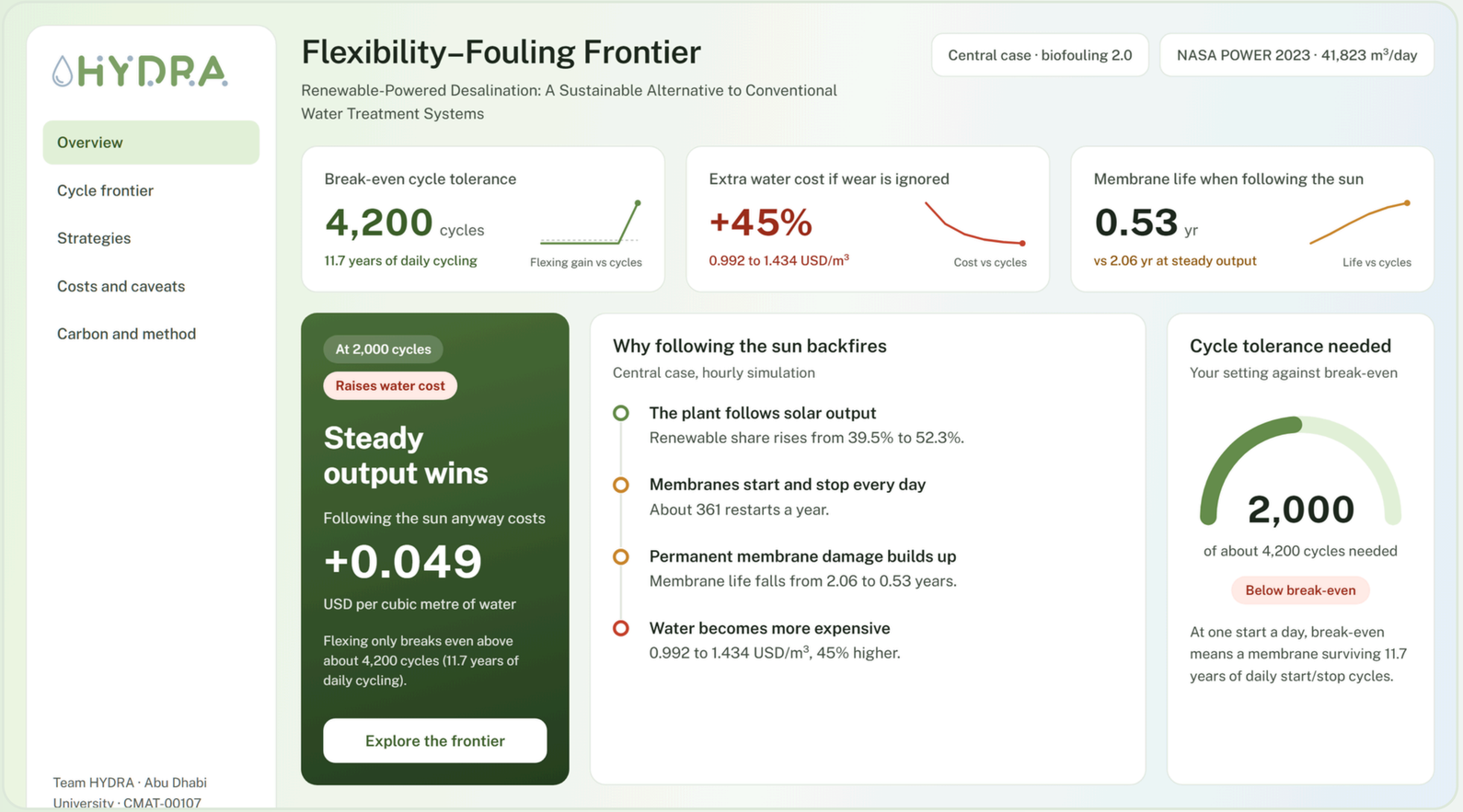
cost of water at 8,000 cycles

198×

The possible loss is 198 times the possible gain. Until cycle tolerance is measured, steady output is the safe choice.

The dashboard lets an operator test the decision

Built from the same simulation results as the report. Runs in a browser, with nothing to install.



1 **Set two inputs**

How heavily the feedwater fouls, and how many start/stop cycles the membrane tolerates.

2 **Read one verdict**

Whether following the sun raises or lowers the cost of water, and the chain of causes behind it.

3 **See the break-even**

How far the membrane is from the roughly 4,200 cycles that flexible operation needs.

Five policy moves, in the right order

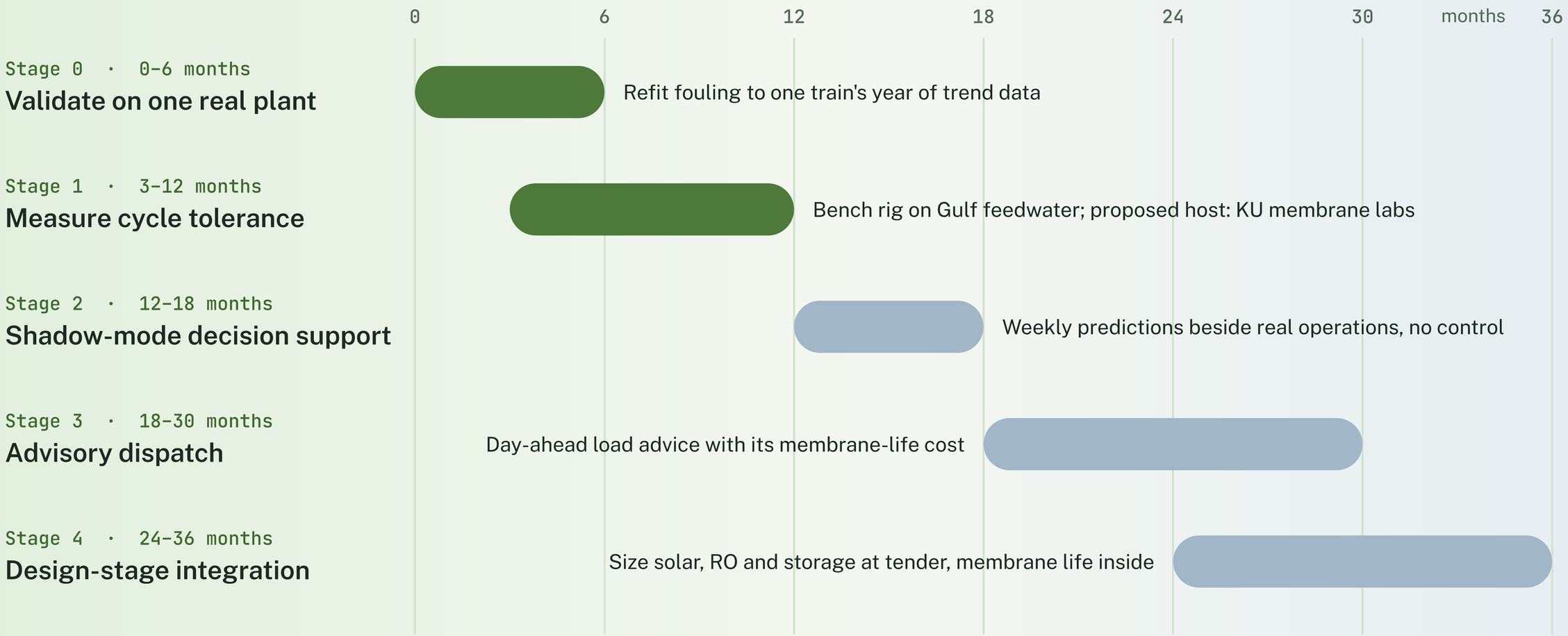
From our UN-aligned policy brief. Each recommendation serves a Theme 1 pillar.

1. **Fund a membrane cycle-tolerance test programme** — Investments for Water
   A few hundred start/stop cycles on Gulf feedwater measures the number the whole result turns on.
2. **Do not count desalination as dispatchable demand** — Water for People
   Until tolerance is measured, treating plants as flexible capacity overstates grid balancing resources.
3. **Introduce time-of-use tariffs only after the measurement** — Water for Prosperity
   At today's membrane capability, strong price spreads reward cycling the membranes cannot support.
4. **Set mandates as a carbon price or absolute emissions** — Investments for Water
   Renewable-percentage targets above about 40% hit the absorption limit and become expensive per tonne.
5. **Direct R&D to cycling-tolerant membranes and restart protocols** — Water for Prosperity
   The avoidable membrane cost, 0.452 USD/m^3, is 22 times the accessible energy saving.

Carbon price for solar to pay — 38 USD/t CO_2 for a 30% renewable share (28–57 across the UAE grid range) · 155 USD/t at the 43% ceiling

From model to plant floor in 36 months

The measurement comes first, because every later stage depends on it.



What we have not shown, and what would settle it

We report conclusions as functions of our assumptions, not as point predictions.

Honest limits

- 34 of 52 parameters are our own assumptions; only 5 come straight from the literature
- Benchmarked against published plant values, not calibrated to operating records
- One site, one resource year (2023), one representative RO train
- **The decisive parameter, permanent loss per start/stop cycle in warm hypersaline seawater, is unmeasured**

What would settle it

One bench experiment

Run two identical elements on Gulf feedwater, one continuously and one with a daily start and stop. Permeability that cleaning cannot recover, plotted against cycle count, gives cycle tolerance directly.

We are asking for

- An operator introduction: one train, one year of normalised trends, cleaning log and start/stop history
- A membrane laboratory to host the bench rig

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Measure the membrane before building the plant around the sun.

One bench experiment decides whether Gulf desalination should follow the sun.

+45%
cost of water when wear is ignored

≈4,200
start/stop cycles to break even

198×
more to lose than to gain

Thank you. Questions welcome.

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Sources quoted on these slides. The technical report holds the full list and a verification log for every number.

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